

Decision tree example w/ gini impurity

Training data:

- features F_1, F_2, F_3
- label/prediction P

Sample #	F_1	F_2	F_3	P
1	T	T	7	F
2	T	F	12	F
3	F	T	18	T
4	F	T	35	T
5	T	T	38	T
6	T	F	50	F
7	F	F	83	F

Gini impurity:

$$G = 1 - \sum_{i=1}^c (p_i)^2$$

c : # of classes

p_i : proportion of data from class i

Start: What goes at the root?

Which feature F_1, F_2, F_3 predicts P the best?

Try: F_1, F_2, F_3

(WA)
 F_1 weighted average of left and right branches

Yes branch: $F_1 = T$

$$G = 1 - \left[\left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}\right)^2 \right] = A$$

No branch: $F_1 = F$

$$G = 1 - \left[\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \right] = B$$

Weighted average:

$$(4 \cdot A + 3 \cdot B) / 7 = 0.405$$

Is F_1 true?

No Yes

Samples: 3, 4, 7 1, 2, 5, 6

F_2

$F_2 = T:$

$$G = 1 - \left[\left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right] = A$$

$F_2 = F:$

$$G = 1 - \left[0 - \left(\frac{3}{3} \right)^2 \right] = B$$

WA:

$$(4 \cdot A + 3 \cdot B) / 7 = 0.214$$

Is F_2 true?

No Yes

Samples: 2, 6, 7 1, 3, 4, 5

ordered

$F_3:$

7
12
18
35
38
50
63

9.5

15

26.5

36.5

44

66.5

Try each threshold:

Is $F_3 \leq 9.5?$

Is $F_3 \leq 15?$

Is $F_3 \leq 26.5?$

Is $F_3 \leq 36.5?$

Is $F_3 \leq 44?$

Is $F_3 \leq 66.5?$

Is $F_3 \leq 36.5?$

No Yes

Samples: 5, 6, 7 1, 2, 3, 4

Yes:

$$G = 1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] = A$$

No:

$$G = 1 - \left[\left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right] = B$$

WA:

$$(4 \cdot A + 3 \cdot B) / 7 = 0.476$$

Each $F_3:$

G

≤ 9.5 0.429

≤ 15 0.343 best (tie)

≤ 26.5 0.476

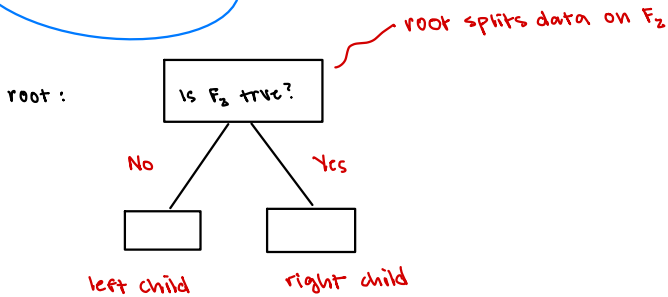
≤ 36.5 0.476

≤ 44 0.343

≤ 66.5 0.426

Pick the smallest WA from F_1, F_2, F_3

F_2 wins!
WA = 0.214



next : repeat with L and R children.

stop when a node is pure